# CENTINEL: A Zero-Cost Cryptographic and Statistical Framework for Real-Time Electoral Audit in Latin America

CENTINEL Project · VectisDev

Independent Research · Honduras · github.com/VectisDev/centinel

**Abstract**

We present CENTINEL (Centinela Electrónico Nacional Técnico Íntegro Neutral Estadístico Libre), an open-source system for real-time cryptographic and statistical auditing of electoral data published by Latin American electoral authorities. CENTINEL captures official JSON feeds, builds a SHA-256 immutable hash chain anchored to the Bitcoin blockchain via OpenTimestamps, and applies 23 forensic detection rules spanning formal statistical tests, arithmetic heuristics, and per-record forensics. A key methodological contribution is the demonstration that first-digit Benford's Law — widely used in election forensics — produces a 100% false positive rate on Honduran electoral count data, while the canonical second-digit formulation (2BL; Mebane, 2006) achieves 1.2% under 500-trial Monte Carlo simulation on clean synthetic data. We present a retroactive forensic analysis of 64 official CNE snapshots from the Honduras 2025 presidential election, identifying five statistically anomalous patterns including a 13.1-hour communication blackout and an act resolution rate of 39.15 acts/minute (3.9× the physically plausible threshold). The system operates at zero infrastructure cost using only GitHub's free tier. All data, code, and analyses are fully reproducible.

**Keywords:** election forensics · Benford's law · cryptographic audit · Honduras · OpenTimestamps · false positive analysis · electoral integrity · Latin America

## 1. Introduction

Electoral authorities in Latin America publish preliminary results through digital transmission systems (TREP) that citizens and observers must accept on trust. No independent technical mechanism has existed to verify, in real time, that published data was not altered post-publication or that statistical patterns are consistent with genuine vote aggregation.

Prior work on election forensics has focused primarily on post-hoc analysis using digit-distribution tests (Mebane, 2006, 2010; Klimek et al., 2012) and turnout-vote share correlations (Deckert et al., 2011). These approaches lack (i) real-time capability, (ii) cryptographic evidence of data integrity, and (iii) systematic false positive calibration against country-specific data.

CENTINEL addresses these gaps with three primary contributions:

**C1.** A zero-cost, reproducible pipeline that captures, cryptographically chains, and forensically analyzes electoral JSON feeds in near real-time (≤5-minute polling interval).

**C2.** Empirical demonstration that first-digit Benford's Law is inappropriate for Honduran electoral count data (100% FP rate), and validation of the canonical second-digit formulation (1.2% FP).

**C3.** A retroactive forensic analysis of the Honduras 2025 presidential election with cryptographically verifiable findings and a publicly archived dataset (DOI: 10.5281/zenodo.20598892).

## 2. System Architecture

CENTINEL is structured in four independent layers that communicate through a shared SQLite state store:

| Layer | Component | Function |
|---|---|---|
| Data Engineering | Poller + Validator | HTTP capture, Pydantic schema validation, normalization |
| Cryptographic | Hash chain + OTS | SHA-256 chaining, Ed25519 signatures, Bitcoin anchoring |
| Detection | Rules engine | 23 forensic rules (Sections A–C), YAML-configured |
| Presentation | Dashboard + PDF | GitHub Pages, executive reports, verify_chain.py |

### 2.1 Cryptographic Integrity Chain

Each captured snapshot $s_i$ is stored with a chained hash $h_i = \text{SHA256}(\text{canonical\_json}(s_i) \| h_{i-1})$. The Merkle root of all snapshots is submitted to public OpenTimestamps calendar servers, which anchor it to the Bitcoin blockchain at zero cost. The standalone verifier *verify_chain.py* requires no dependencies beyond Python 3.6 standard library, enabling offline verification by any third party.

### 2.2 Detection Rules

The 23 rules are grouped into three families:

**Section A — Formal statistical tests (10 rules):** Canonical second-digit Benford (2BL), last-digit uniformity (chi-square), Wald-Wolfowitz runs test, Pearson correlation of turnout vs. vote share, geographic coefficient of variation, Law of Large Numbers convergence Z-score, historical participation Z-score, Isolation Forest anomaly detection, and granular multi-subtest rule.

**Section B — Arithmetic and heuristic rules (10 rules):** Basic consistency, null/blank vote thresholds, impossible turnout, snapshot jump detection, processing speed, participation anomaly, trend deviation, statistical irreversibility, table consistency, duplicate/missing mesa detection.

**Section C — Per-record JSON forensics (3 rules):** Record mutation detection via per-mesa SHA-256 fingerprints, arithmetic impossibility per record, and late-batch mesa injection.

## 3. Statistical Methodology and False Positive Calibration

### 3.1 The Benford's Law Problem in Electoral Data

First-digit Benford's Law (BL1) states $P(d) = \log_{10}(1 + 1/d)$ for $d \in \{1,...,9\}$. Its application to electoral forensics has been widely criticized (Mebane, 2011; Deckert et al., 2011) because vote counts by polling station typically span less than one order of magnitude, violating the scale-invariance assumption.

We demonstrate this empirically: BL1 applied to 500 synthetic samples drawn from Lognormal($\mu$=5.5, $\sigma$=1.8) — calibrated to Honduran per-mesa distributions (200–500 votes) — triggers **in 100% of trials** at the standard chi-square threshold ($p < 0.05$ OR MAD > 0.006). This renders BL1 statistically indefensible as an operational alert.

### 3.2 Canonical Second-Digit Benford (2BL)

The second-digit formulation (Mebane, 2006) defines $P(d_2 = k) = \sum_{d_1=1}^{9} \log_{10}(1 + 1/(10 \cdot d_1 + k))$ for $k \in \{0,...,9\}$. It does not require scale invariance and is robust to the limited range of per-mesa vote counts. With dual confirmation (chi-square $p < 0.01$ AND MAD > 0.012 for CRITICAL severity), we observe a false positive rate of **1.2%** (6/500 trials), within the design threshold of 5%.

### 3.3 Unified Z-Score Conventions

CENTINEL dev-v10 used three inconsistent Z-score formulations. dev-v12 unifies them into two canonical families:

| Family | Formula | Application | Thresholds |
|---|---|---|---|
| A — Proportion | Z = (p■ – p■) / √(p■(1–p■)/n) | Vote shares, turnout | W: 2.576 · C: 3.291 |
| B — Empirical | Z = (x – x■) / s [ddof=1] | Counts, deltas, historical | W: 2.576 · C: 3.291 |

The Bessel correction (ddof=1) is mandatory for finite samples (n < 50, typical for Honduras' 18 departments). The population estimator (ddof=0) systematically underestimates σ and inflates Z-scores, increasing false positives.

### 3.4 Monte Carlo Validation Results

| Rule / Component | FP Rate dev-v10 | FP Rate dev-v12 | Change |
|---|---|---|---|
| Benford 1st digit (BL1) | 100.0% (500/500) | 0% (INFO only) | –100 pp |
| Benford 2nd digit 2BL — CRITICAL | — | 1.2% (6/500) | new |
| Z-score empirical WARNING | ~3.5% (est.) | 2.4% (12/500) | –1.1 pp |
| Z-score empirical CRITICAL | ~0.8% (est.) | 0.6% (3/500) | –0.2 pp |
| Z-score proportion WARNING/CRITICAL | ~1%/~0.1% | 0.0% (0/500) | –1 pp |

Table 1. Monte Carlo results (N=500, seed=2025, Lognormal(5.5, 1.8) clean data). W=Warning, C=Critical.

## 4. Honduras 2025 — Retroactive Forensic Analysis

We applied CENTINEL to 64 official CNE JSON snapshots captured manually between December 3–10, 2025, covering the count of the November 30, 2025 Honduran general elections. The dataset is publicly archived at DOI 10.5281/zenodo.20598892 with a Bitcoin timestamp proof.

### 4.1 Operational Parameters

| Parameter | Value |
|---|---|
| Snapshots analyzed | 64 |
| Period covered | Dec 3–10, 2025 (168.6 hours) |
| Capture method | Manual (not automated; ≤5-min system not yet deployed) |
| Final scrutiny | 99.4% (19,052 / 19,167 actas) |
| Inconsistent actas at closure | 2,773 (14.6% of disclosed) |
| Median inter-snapshot interval | 60 minutes |

### 4.2 Statistical Findings

| Finding | Rule | Metric | Threshold |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Communication blackout | snapshot_jump | 13.1 h gap (Dec 3 22:00→Dec 4 11:06) | > 10 min gap |
| Impossible resolution rate | processing_speed | 39.15 actas/min | ~10/min plausible |
| Prolonged stagnation | irreversibility | 25 consecutive cycles, inconsistentes frozen | ≥ 4 cycles |
| Progressive injection | participation_anomaly | 17 low-delta cycles (Δ<100) with backlog >1,000 | Δ threshold |
| Overnight blackouts | snapshot_jump | 7 events, 11–25 h each, systematic pattern | > 10 h gap |

Table 2. Forensic findings from HN 2025 retroactive analysis. All findings are statistical anomalies in published CNE data; no claim of electoral fraud is made.

### 4.3 Cryptographic Integrity

The Merkle root of all 64 snapshots (f17dbfe188d2f1a1b31248a343ae3397984ef2ec6550bd1bcded03d31d2a7780) was submitted to the OpenTimestamps Bitcoin calendar. The resulting proof file (MERKLE_ROOT_HN2025.ots) is included in the Zenodo dataset. Any third party can verify: (i) the hash chain integrity using verify_chain.py with no external dependencies, and (ii) the Bitcoin temporal anchor using the opentimestamps-client library.

## 5. Threshold Calibration Against Real Data

Dev-v10 thresholds were largely arbitrary. We calibrated them empirically against the 63 valid HN 2025 snapshots:

| Rule | Parameter | dev-v10 | dev-v12 Calibrated | Empirical Basis |
|---|---|---|---|---|
| R8/R13 | min_turnout_pct | 40% | 70% | Observed p1=80%; buffer 10 pp |
| R8/R13 | max_turnout_pct | 90% | 100% | Observed max=99.4% |
| R5 | critical_r (Pearson) | 0.85 | 0.85 ✓ | r=0.76 in HN2025 — no FP |
| R5 | warning_r (new) | — | 0.70 | HN2025 r=0.76 warrants early warning |
| R6 | critical_cv | 0.45 | 0.45 ✓ | CV=0.019 in synthetic depts |
| R6 | warning_cv (new) | — | 0.15 | 8× buffer over observed CV |
| R14 | max_change_pct | 5% | 5% ✓ | p99 of valid intervals = 2.81% |

Table 3. Threshold calibration summary.

## 6. Discussion

### 6.1 Limitations

The HN 2025 analysis is retroactive and based on manual captures with a 60-minute median interval, not the ≤5-minute automated polling for which CENTINEL is designed. Departmental-level data was unavailable; CNE publishes national aggregates only during scrutiny, and historical TREP data from prior elections (2013, 2017, 2021) is not publicly archived in structured format. The synthetic departmental data used for CV calibration is generated from national totals and should not be treated as empirically representative.

### 6.2 Interpretation of Findings

The five anomalous patterns identified in Section 4 are statistical signals in published data. Their causes — technical failures, institutional procedures, or intentional manipulation — cannot be determined from data analysis alone. CENTINEL makes no claim regarding electoral fraud.

### 6.3 Generalizability

The system is configured for six LATAM countries (Honduras, Guatemala, El Salvador, Nicaragua, Mexico, Colombia). Adding a new country requires a single configuration file specifying the electoral authority's JSON schema and endpoint. Zero-cost operation (GitHub free tier) ensures accessibility without institutional support.

## 7. Conclusion

CENTINEL demonstrates that cryptographically-verified, statistically-rigorous electoral audit is achievable at zero infrastructure cost. The system's primary methodological contribution — empirical falsification of first-digit Benford's Law as an operational election forensics tool, and validation of the canonical 2BL formulation — has direct implications for the broader election forensics literature.

The Honduras 2025 retroactive analysis provides a reproducible, cryptographically anchored reference case for the application of these methods to real electoral data. All code, data, and analyses are publicly available under AGPL-3.0.

## Acknowledgments

Academic validation in progress with Prof. Devis Alvarado, Department of Mathematics and Statistics, Universidad Pedagógica Nacional Francisco Morazán (UPNFM), Tegucigalpa, Honduras.

## References

CENTINEL Project (2026). Honduras 2025 Presidential Election — CNE TREP Snapshots (64 JSON files) [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.20598892

Deckert, J., Myagkov, M., & Ordeshook, P. C. (2011). Benford's Law and the detection of election fraud. *Political Analysis, 19*(3), 245–268. https://doi.org/10.1093/pan/mpr014

Klimek, P., Yegorov, Y., Hanel, R., & Thurner, S. (2012). Statistical detection of systematic election irregularities. *Proceedings of the National Academy of Sciences, 109*(41), 16469–16473. https://doi.org/10.1073/pnas.1210722109

Mebane, W. R., Jr. (2006). *Election forensics: Vote counts and Benford's law*. Working paper, University of Michigan.

Mebane, W. R., Jr. (2010). Fraud in the 2009 presidential election in Iran? *Chance, 23*(1), 6–15. https://doi.org/10.1080/09332480.2010.10739785

Mebane, W. R., Jr. (2011). Comment on Benford's Law and the detection of election fraud. *Political Analysis, 19*(3), 269–272. https://doi.org/10.1093/pan/mpr024

NIST/SEMATECH (2012). *NIST/SEMATECH e-Handbook of Statistical Methods*. https://doi.org/10.18434/M32189

OpenTimestamps (2016). *OpenTimestamps: Scalable, trust-minimized, distributed timestamping with Bitcoin*. https://opentimestamps.org